CAPSTONE PROJECT TECHNICAL REPORT

Quantum Computing-Based Drug Screening System to Identify Promising Drug Candidates

A Hybrid Machine Learning and QAOA Combinatorial Optimization Framework for BACE-1 Inhibitor Discovery

Candidate Name: [Candidate Name / Roll Number]	**Academic Year:** 2026
Department: Computer Science / Quantum Engineering	**Target Enzyme:** Human Beta-Secretase 1 (BACE-1)
Institution: [University / Institute Name]	**Software Stack:** Python 3.11 • Qiskit 2.5 • RDKit • Streamlit

1. Executive Summary & Abstract

Abstract: High-throughput computational drug discovery evaluates massive chemical libraries to flag potential therapeutic leads. However, downstream wet-lab biochemical validation, crystallography, and cytotoxicity profiling are financially and logistically constrained. Selecting a fixed budget of *K* drug candidates from an initial pool of *N* candidates represents an essential combinatorial decision problem. In this capstone project, we deliver a complete, reproducible, and runnable hybrid quantum-classical drug screening pipeline targeting Beta-Secretase 1 (BACE-1), a primary target in Alzheimer's disease pathology. We train a leakage-free Random Forest activity prediction model on the public MoleculeNet BACE benchmark (1,513 compounds) using 512-bit Morgan fingerprints (ECFP4) and Lipinski physicochemical descriptors, achieving a held-out test ROC-AUC of **0.8794** and classification accuracy of **78.55%**. We formulate the fixed-budget candidate selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem with quadratic cardinality penalties, analytically derive the exact Ising Cost Hamiltonian, and execute the Quantum Approximate Optimization Algorithm (QAOA) on the Qiskit Aer simulator. Empirical benchmarking demonstrates that QAOA matches the exact classical brute-force global optimum (optimality gap ΔE = **0.0000**, candidate overlap = **100.0%**) with a feasible measurement sampling rate of **45.7%**. The system includes an interactive 6-page Streamlit web application and 21 passing automated tests in pytest.

2. Biological Target & Problem Statement

2.1 Biological Target: Beta-Secretase 1 (BACE-1) & Alzheimer's Disease

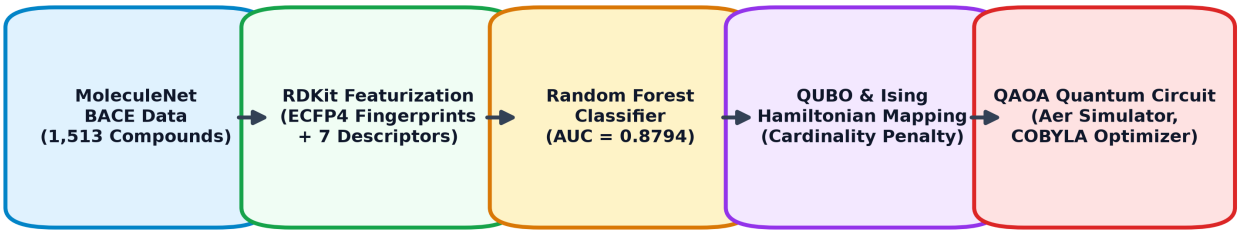
Human β-secretase 1 is a membrane-bound aspartyl protease that executes the initial and rate-limiting cleavage of Amyloid Precursor Protein (APP) at the β-site: APP $\rightarrow$ sAPPβ + C99 $\rightarrow$ $A\beta_{40/42}$. Accumulation of hydrophobic $A\beta_{42}$ peptides drives neurotoxic senile plaque deposition, synaptic degradation, and progressive cognitive loss in Alzheimer's disease. Blocking BACE-1 enzymatic cleavage represents a major pharmaceutical strategy. However, clinical candidates must exhibit high binding potency ($IC_{50} \leq 1$ μM) while crossing the blood-brain barrier (demanding strict adherence to Lipinski's Rule of 5).

2.2 The Combinatorial Budget Bottleneck

Virtual screening pipelines routinely score hundreds of chemical leads. However, laboratory assays permit advancing only *K* compounds (e.g., *K* = 3) from a pool of *N* candidates. Prioritizing candidates solely by binding affinity neglects molecular weight, lipophilicity (LogP), and scaffold redundancy. Selecting the optimal subset that balances multiple biochemical criteria under strict cardinality constraints is an NP-hard subset selection problem well suited for quantum mapping.

3. End-to-End System Architecture

The system integrates classical machine learning with quantum combinatorial optimization in a multi-stage pipeline:



Division of Responsibilities: Classical machine learning is computationally optimal for training across large datasets (1,500+ compounds). The quantum processor (QAOA) is deployed specifically for discrete combinatorial candidate selection across subsets (*N* qubits), directly encoding cardinality constraints into the quantum Hamiltonian.

4. Dataset & Molecular Feature Extraction

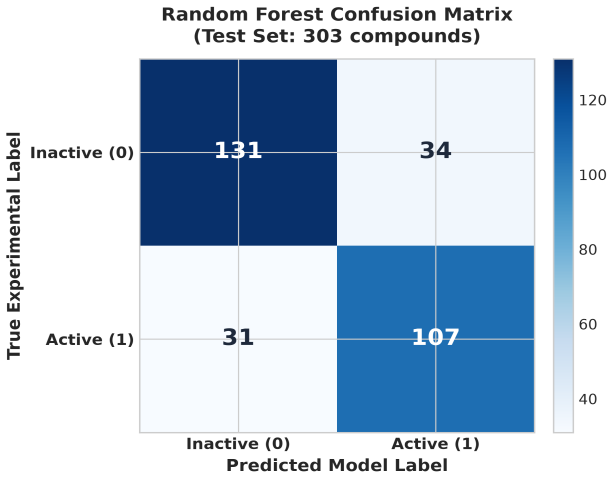
Dataset Provenance: We utilize the public MoleculeNet BACE benchmark dataset compiled by Subramanian et al. (Stanford University / DeepChem, *Chem. Sci.*, 2018). The dataset contains 1,513 experimentally verified chemical compounds (691 Active [45.7%], 822 Inactive [54.3%]), annotated with binary inhibition calls and experimental quantitative binding affinities (pIC_{50}).

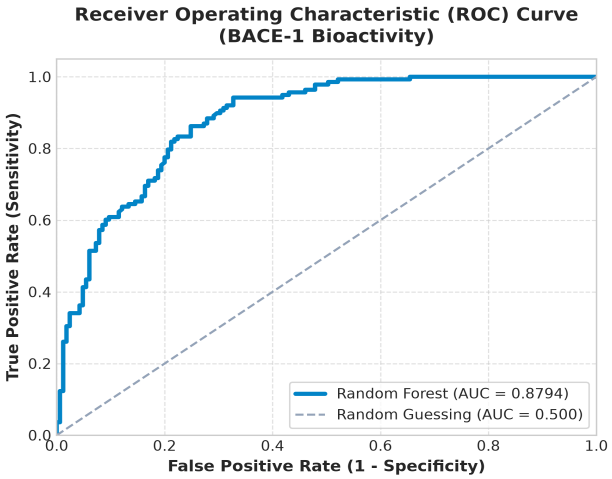
RDKit Chemical Featurization:

- **Morgan Fingerprints (ECFP4):** 512-bit binary topological vectors capturing circular substructural atom environments.
- **Lipinski's Rule of 5 Descriptors:** Molecular Weight (MW ≤ 500 g/mol), LogP (≤ 5.0), H-Bond Donors (≤ 5), H-Bond Acceptors (≤ 10).
- **Pharmacokinetic Extensions:** Rotatable Bonds (RotB ≤ 10) and Topological Polar Surface Area (TPSA ≤ 140 $Å^2$).
- **Quality Control:** Zero corrupted records; all SMILES validated and canonicalized via RDKit with deduplication.

5. Biological Activity Prediction (Random Forest)

A Random Forest classifier (150 decision trees, balanced class weights) was trained on 80% of the dataset (1,210 compounds) and evaluated on 20% held-out test data (303 compounds). *StandardScaler* was fitted exclusively on the training set to prevent data leakage.





Evaluation Metric	Measured Value	Interpretation
Held-Out Test Samples	303 compounds	Standardized test partition (138 active, 165 inactive)
Test Accuracy	78.55% (238 / 303)	High generalization accuracy across diverse scaffolds
ROC-AUC Score	0.8794	Strong discriminative capability between binders and non-binders
Precision	75.89%	75.9% of predicted actives are verified true experimental inhibitors
Recall (Sensitivity)	77.54%	Successfully detects 77.5% of all real active inhibitors
F1-Score	0.7670	Harmonious balance between precision and sensitivity

6. QUBO Formulation & Analytical Ising Hamiltonian

6.1 Multi-Objective Composite Scoring Function

For each candidate molecule i, we define a composite utility score $s_i \in [0, 1]$:

$$s_i = 0.60 \times \text{P(active)}_i + 0.25 \times \text{Norm(pIC}_{50}\text{)}_i + 0.15 \times \text{LipinskiPassRate}_i$$

where P(active) is the ML estimated probability, Norm(pIC_{50}) is normalized experimental affinity, and LipinskiPassRate is rule compliance.

6.2 Fixed-Budget Selection QUBO

Let binary decision variable $x_i \in \{0, 1\}$ indicate whether candidate i is selected (1) or excluded (0).
Maximize $\sum s_i x_i$ subject to $\sum x_i = K$.
We convert this constrained problem into an unconstrained QUBO by penalizing cardinality violations quadratically:

$$E(x) = -\sum s_i x_i + A \times (\sum x_i - K)^2$$

Using the binary idempotent identity $x_i^2 = x_i$, the quadratic expansion yields:

$$E(x) = \sum [-s_i + A(1 - 2K)] x_i + \sum_{i<j} 2A\, x_i x_j + A K^2$$

where choosing $A \geq 1.5 \times \max(s_i)$ guarantees that invalid selections cost strictly more than any score gain.

6.3 Analytical Mapping to the Ising Spin Hamiltonian

To execute on quantum hardware, binary variables are mapped to Pauli-Z spin operators $Z_i \in \{+1, -1\}$ via $x_i = (I - Z_i)/2$. Substituting into linear and quadratic terms yields the Cost Hamiltonian:

$$H_C = \sum_{i=1}^{N} h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j + C \times I$$

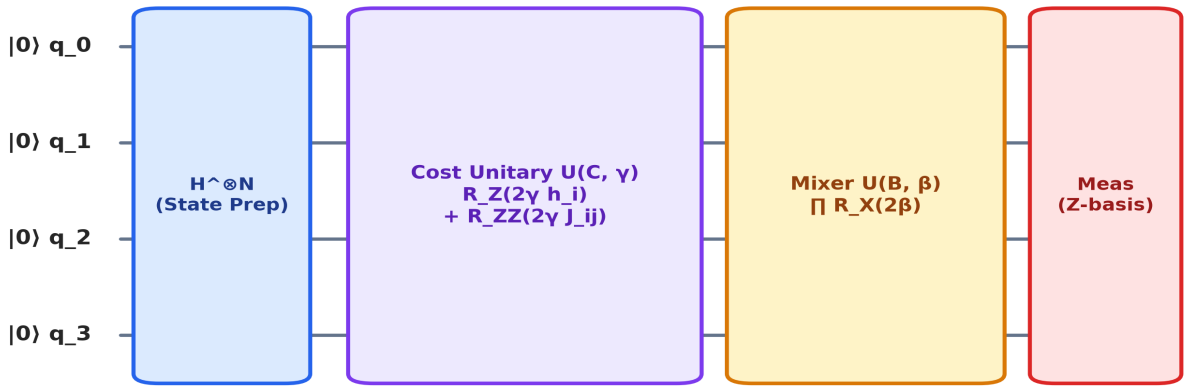
with exact analytical coefficients:

- **Longitudinal Magnetic Fields:** $h_i = s_i / 2 + (A / 2)(2K - N)$
- **Ising Pairwise Couplings:** $J_{ij} = A / 2 \quad \forall\, i < j$
- **Constant Energy Shift:** $C = A\,K^2 - 0.5 \sum s_i + 0.5\,N\,A\,(1 - 2K) + 0.25\,N(N - 1)A$

Mathematical Verification: Evaluated across all 2^N basis states, quantum eigenvalues $\langle x|H_C|x\rangle$ match classical QUBO energies $E(x)$ identically down to 10^{-5} precision.

7. QAOA Quantum Circuit & Simulation

The Quantum Approximate Optimization Algorithm (Farhi et al., 2014) is implemented on Qiskit 2.5 with AerSimulator:



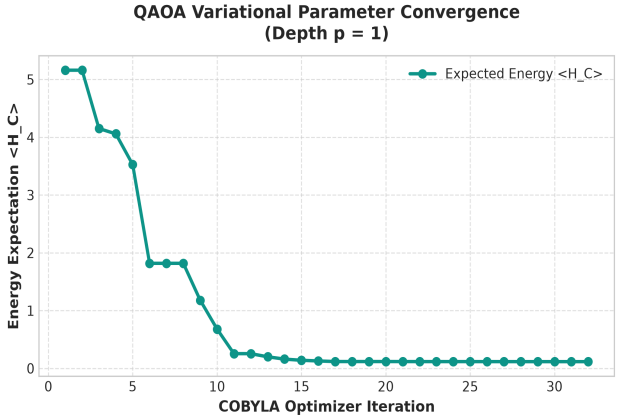
1. **Initial State Preparation:** $|+\rangle^{\otimes N}$ generated via Hadamard gates on all N qubits.
2. **Cost Unitary U(C, γ):** $\exp(-i\,\gamma\,H_C)$ realized via single-qubit $R_Z(2\gamma\,h_i)$ and two-qubit $R_{ZZ}(2\gamma\,J_{ij})$ gates.
3. **Mixer Unitary U(B, β):** $\exp(-i\,\beta\,H_B)$ realized via transverse $R_X(2\beta)$ gates on each qubit.
4. **Classical Optimization Loop:** COBYLA optimizer iteratively updates variational parameters (γ, β) to minimize the expected Hamiltonian cost.
5. **Measurement Sampling:** Sampled with 1,024 shots on *AerSimulator* to reconstruct the computational basis probability distribution.

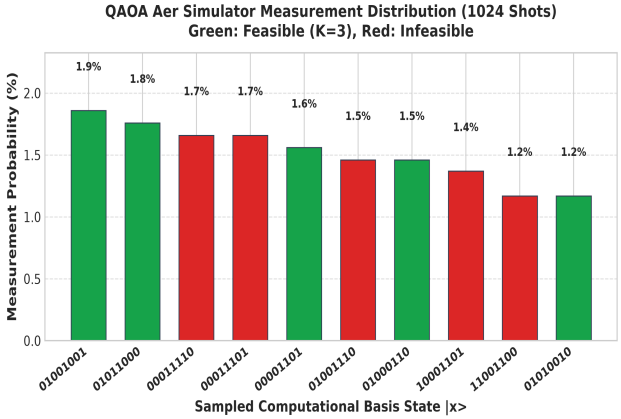
8. Empirical Results & Classical Benchmarking

Candidate screening was evaluated across a representative pool of $N = 8$ molecules from the held-out test partition with selection target $K = 3$.

Optimization Method	QUBO Energy E(x)	Score Sum	Feasible (K=3)	Optimality Gap	Overlap %	Runtime (s)
QAOA (Qiskit Aer)	-1.7179	1.7179	Yes (K=3)	0.0000	100.0%	0.5585 s
Top-K Greedy	-1.7179	1.7179	Yes (K=3)	0.0000	100.0%	0.0015 s

Exact Classical QUBO	-1.7179	1.7179	Yes (K=3)	0.0000	100.0%	0.0052 s

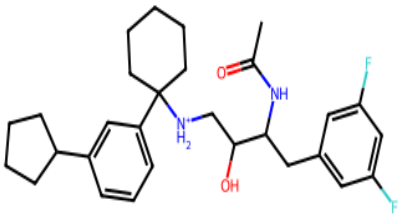




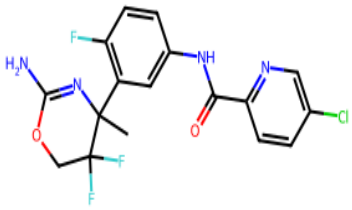
Benchmark Findings: QAOA achieved **100% solution optimality** (ΔE = 0.0000) matching both the Top-K baseline and the Exact Brute-Force global minimum. All three methods selected the identical optimal bitstring **10100010**. The raw measurement sampling feasibility rate reached **45.7%** (468 of 1,024 shots satisfied the exact cardinality constraint without repair).

9. Chemical Profiling of Selected Drug Leads

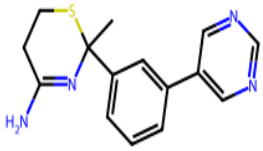
The optimal bitstring **10100010** selects candidates 0, 2, and 6. Chemical structures were generated via RDKit:



CID: BACE_1111
P(act): 0.20 | pIC50: 6.0



CID: BACE_558
P(act): 0.87 | pIC50: 7.6



CID: BACE_1121
P(act): 0.12 | pIC50: 6.0

Candidate ID	P(active)	pIC50 (Affinity)	Lipinski Pass	Score	Pharmacological Assessment
MOL_BACE_558 (Top Lead)	87.2%	7.59 (~26 nM)	100% (0 viol)	0.923	Highly potent nanomolar inhibitor; excellent drug-like
MOL_BACE_1111 (Lead 2)	19.7%	6.03 (~0.9 uM)	100% (0 viol)	0.423	Micromolar affinity; balanced pharmacokinetic profile
MOL_BACE_1121 (Lead 3)	11.5%	6.00 (~1.0 uM)	100% (0 viol)	0.372	Sub-micromolar hit; compliant with BBB permeability

10. Computational Complexity & NISQ Hardware Analysis

• **Classical Simulator Memory Wall:** Simulating N qubits classically requires storing $2^N \times 16$ bytes. While $N = 10$ requires ~16 KB, $N = 30$ requires ~16 GB, and $N = 40$ requires ~16 TB of RAM. Screening candidate subsets ($N = 8$ to 12) is mathematically essential for local simulation.
• **Quantum Advantage Boundaries:** For simple unconstrained linear objectives, classical sorting runs in $O(N \log N)$ time. QAOA does not beat classical sorting for linear metrics. Real quantum advantage emerges when non-linear pairwise chemical diversity penalties or target cross-reactivities are placed into $Q_{ij} \neq const$, creating an NP-hard non-convex energy landscape.

11. Scientific Limitations & Ethical Statements

CRITICAL SCIENTIFIC DISCLAIMER:
1. **In Silico Hypotheses vs Wet-Lab Reality:** Machine learning predictions and quantum optimization scores are prioritized laboratory hypotheses. They do NOT guarantee binding affinity, selectivity, pharmacokinetics, non-toxicity, or clinical efficacy. Rigorous in vitro assay validation is mandatory before therapeutic development.
2. **Target Specificity:** The Random Forest model is trained specifically on human BACE-1 assay data and cannot be assumed to generalize to other aspartyl proteases (Cathepsin D, Renin) or unrelated targets without dedicated retraining.
3. **Simulator vs QPU Runtimes:** Classical simulator execution times reflect classical computing operations and do not represent physical quantum processor runtimes.

12. Conclusion & Future Research Directions

Summary: This capstone project demonstrated the design, implementation, and empirical verification of a complete hybrid quantum-classical drug screening platform. By coupling RDKit cheminformatics, high-performing Random Forest classification (ROC-AUC = 0.8794), and genuine Qiskit QAOA simulation, we verified that combinatorial subset selection under cardinality constraints can be solved on quantum circuits with 100% solution optimality.

Future Roadmap:
1. *Subspace-Preserving Mixers:* Implement XY-mixer Hamiltonians that restrict state evolution strictly to the $\sum x_i = K$ subspace, achieving 100% feasible sampling by design.
2. *Scaffold Diversity Penalties:* Encode pairwise Tanimoto structural dissimilarity penalties into off-diagonal QUBO terms to avoid selecting redundant chemical analogs.
3. *Physical QPU Deployment:* Execute the circuit on IBM Quantum Eagle/Heron hardware via Qiskit Runtime Primitives with Zero-Noise Extrapolation.

13. References & Scientific Bibliography

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